

Introduction to Direct Access Files.

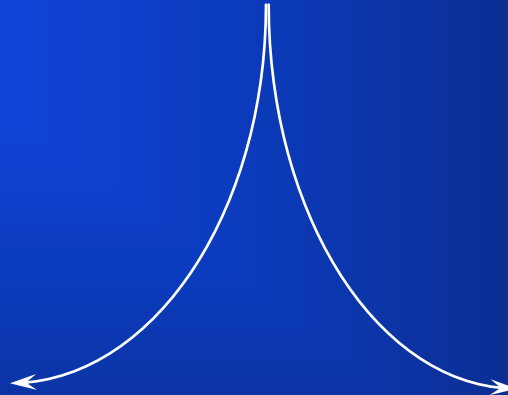
Sequential Files - Adding a Record

Ordered

Rec001
Rec005
Rec045
Rec090
Rec100
Rec150
Rec300
Rec325

^Z

Rec085



Unordered

Rec300
Rec150
Rec005
Rec090
Rec045
Rec100
Rec001
Rec325

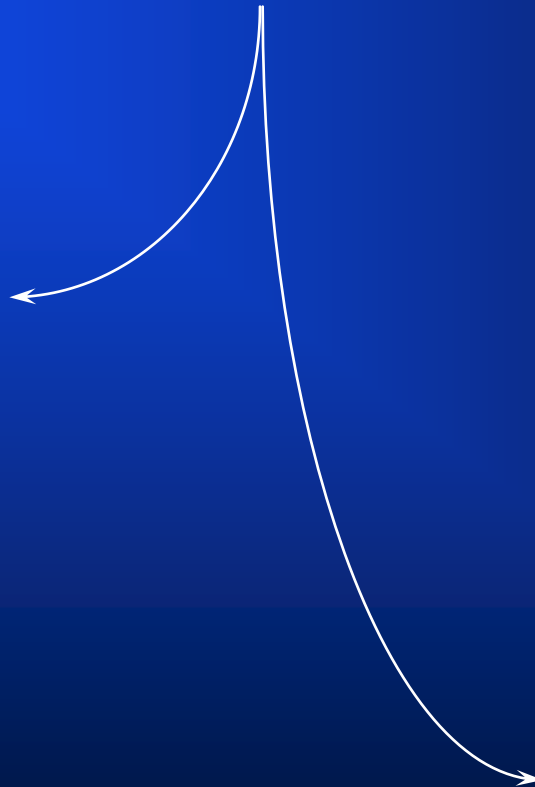
^Z

Sequential Files - Adding a Record

New-Ordered

Rec001
Rec005
Rec045
Rec085
Rec090
Rec100
Rec150
Rec300
Rec325
^Z

Rec085



Extend-Unordered

Rec300
Rec150
Rec005
Rec090
Rec045
Rec100
Rec001
Rec325
Rec085
^Z

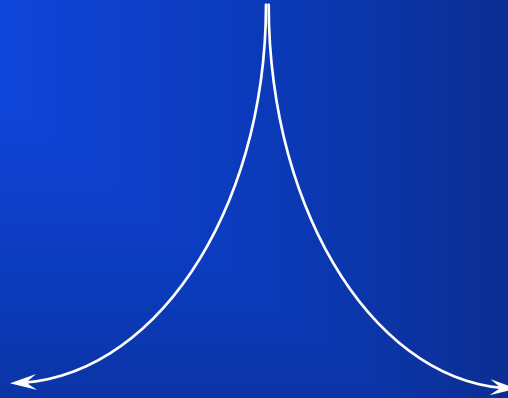
Sequential Files - Deleting a Record

Ordered

Rec001
Rec005
Rec045
Rec090
Rec100
Rec150
Rec300
Rec325

^Z

Rec150

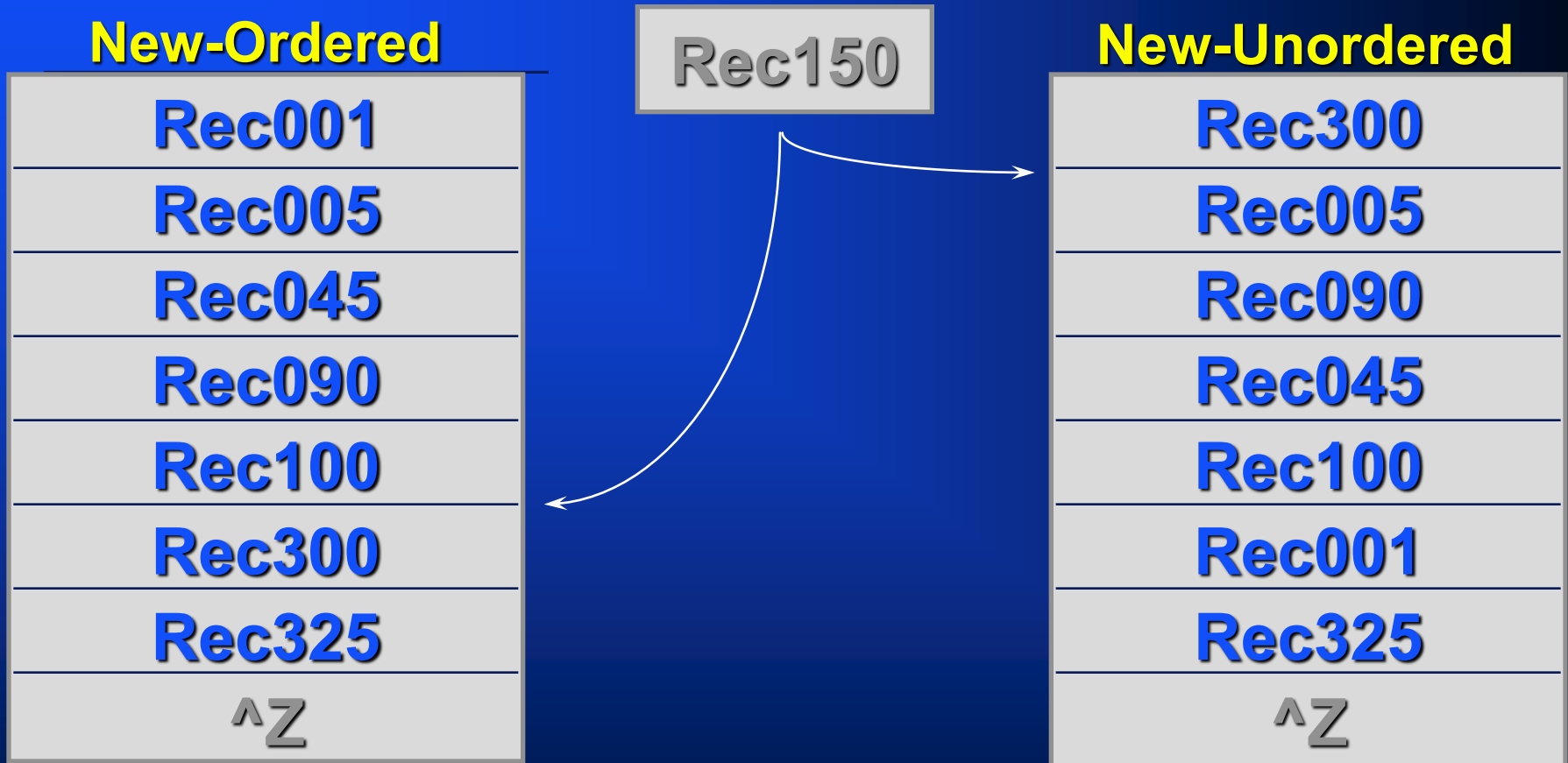


Unordered

Rec300
Rec150
Rec005
Rec090
Rec045
Rec100
Rec001
Rec325

^Z

Sequential Files - Deleting a Record



Sequential Files - Amending a Record

Ordered

Rec001

Rec005

Rec045

Rec090

Rec100

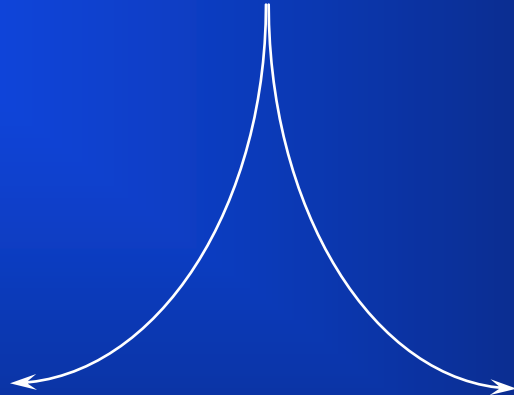
Rec150

Rec300

Rec325

^Z

Rec045



Unordered

Rec300

Rec150

Rec005

Rec090

Rec045

Rec100

Rec001

Rec325

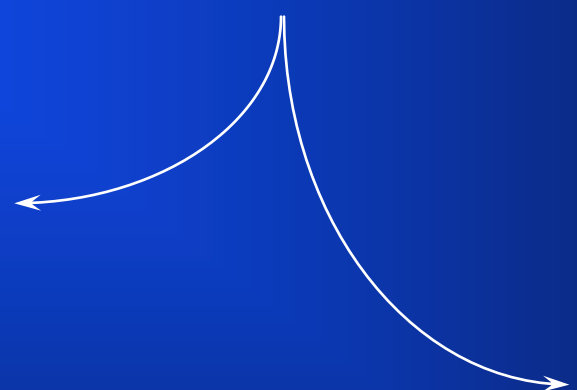
^Z

Sequential Files - Amending a Record

New-Ordered

Rec001
Rec005
Rec045
Rec090
Rec100
Rec150
Rec300
Rec325
^Z

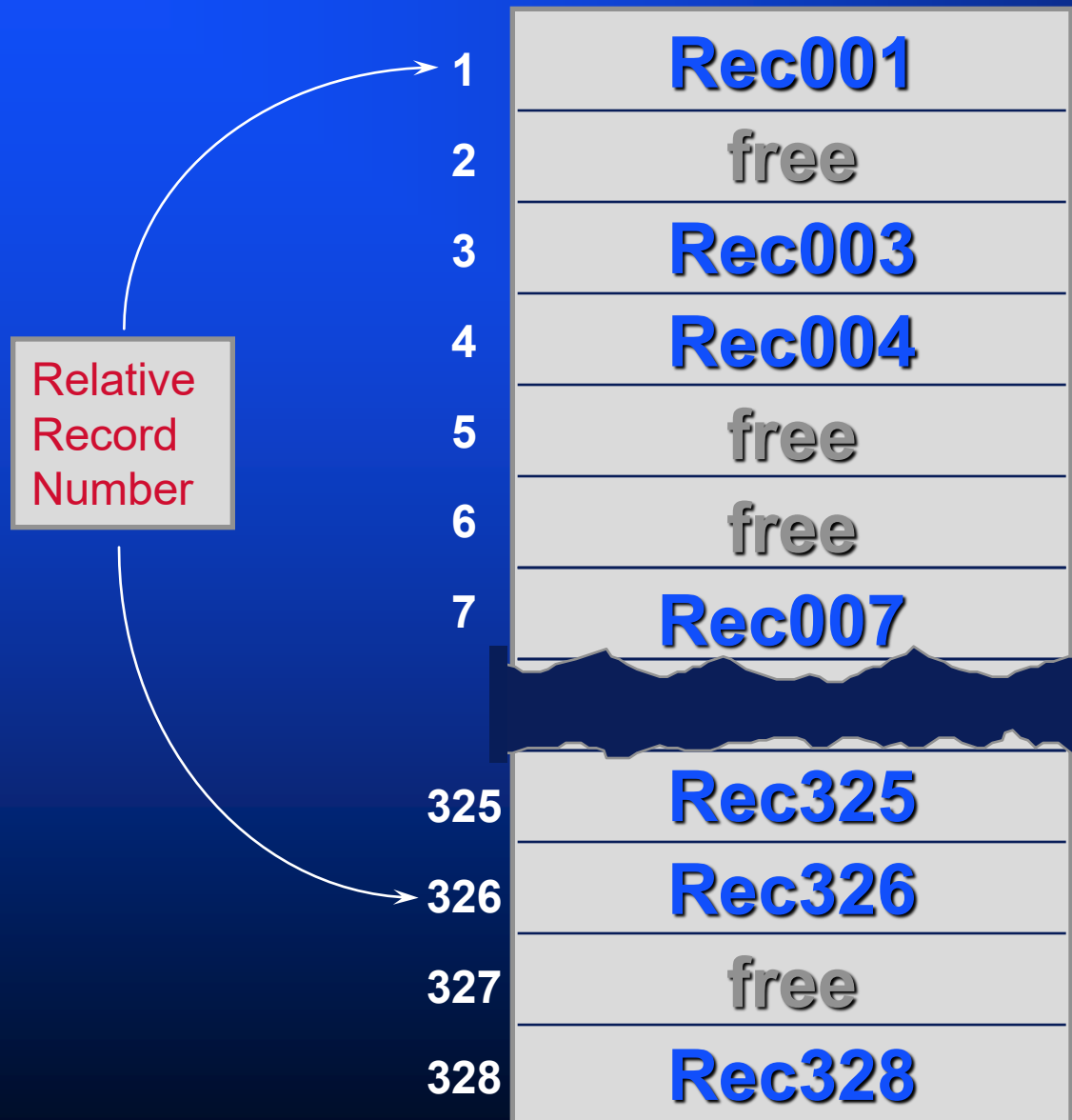
Rec045



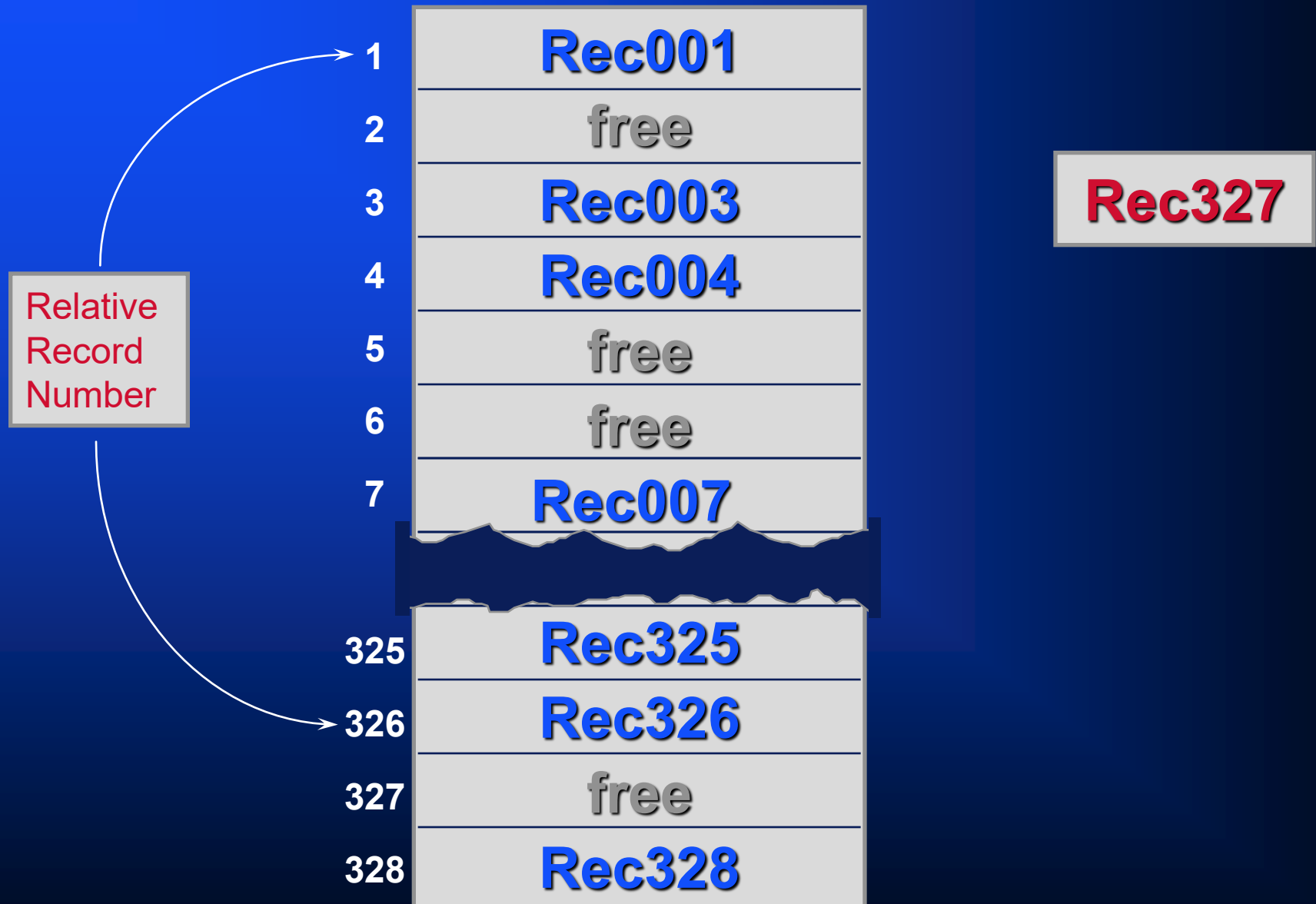
New-Unordered

Rec300
Rec150
Rec005
Rec090
Rec045
Rec100
Rec001
Rec325
^Z

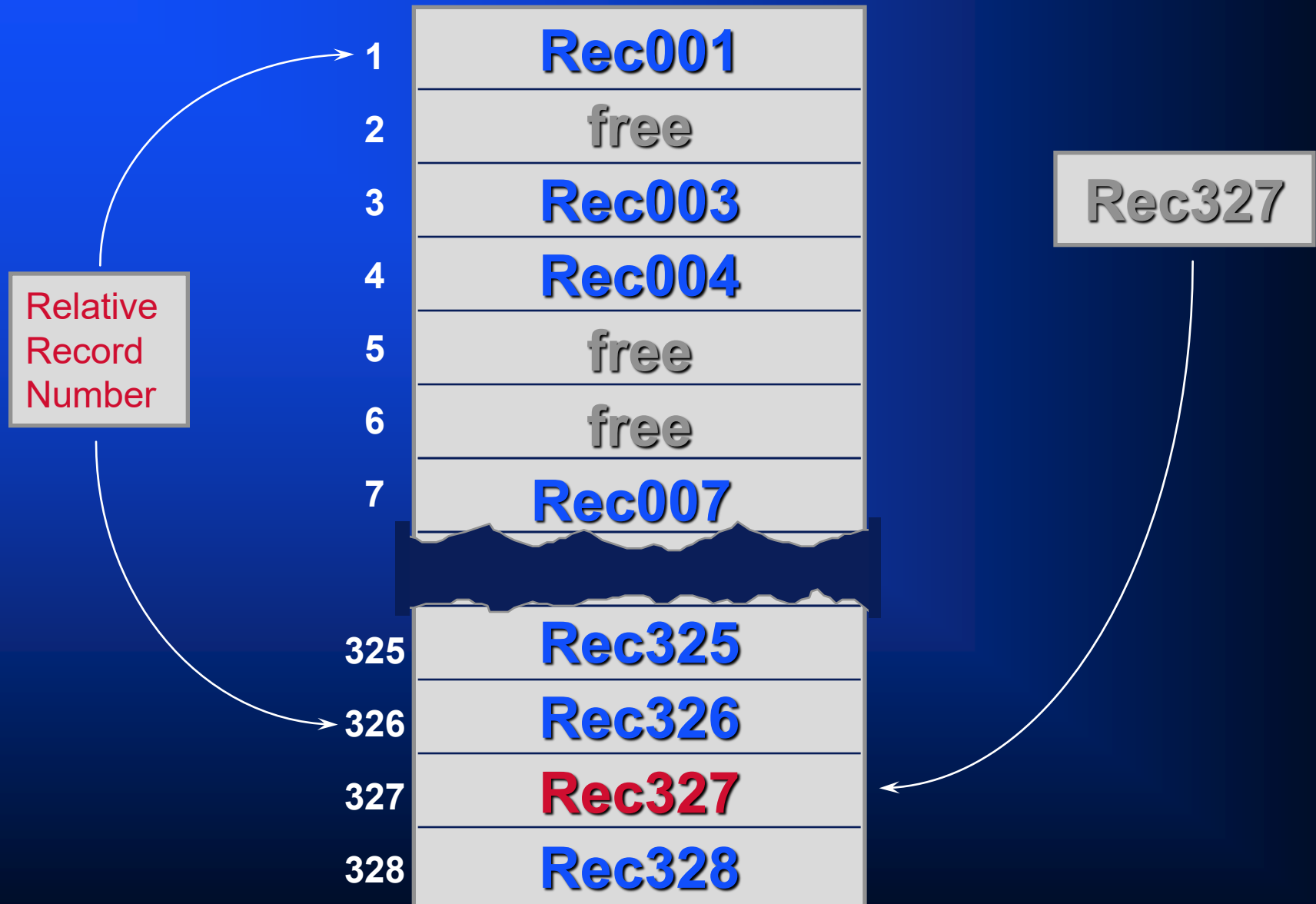
Relative Files - Organization



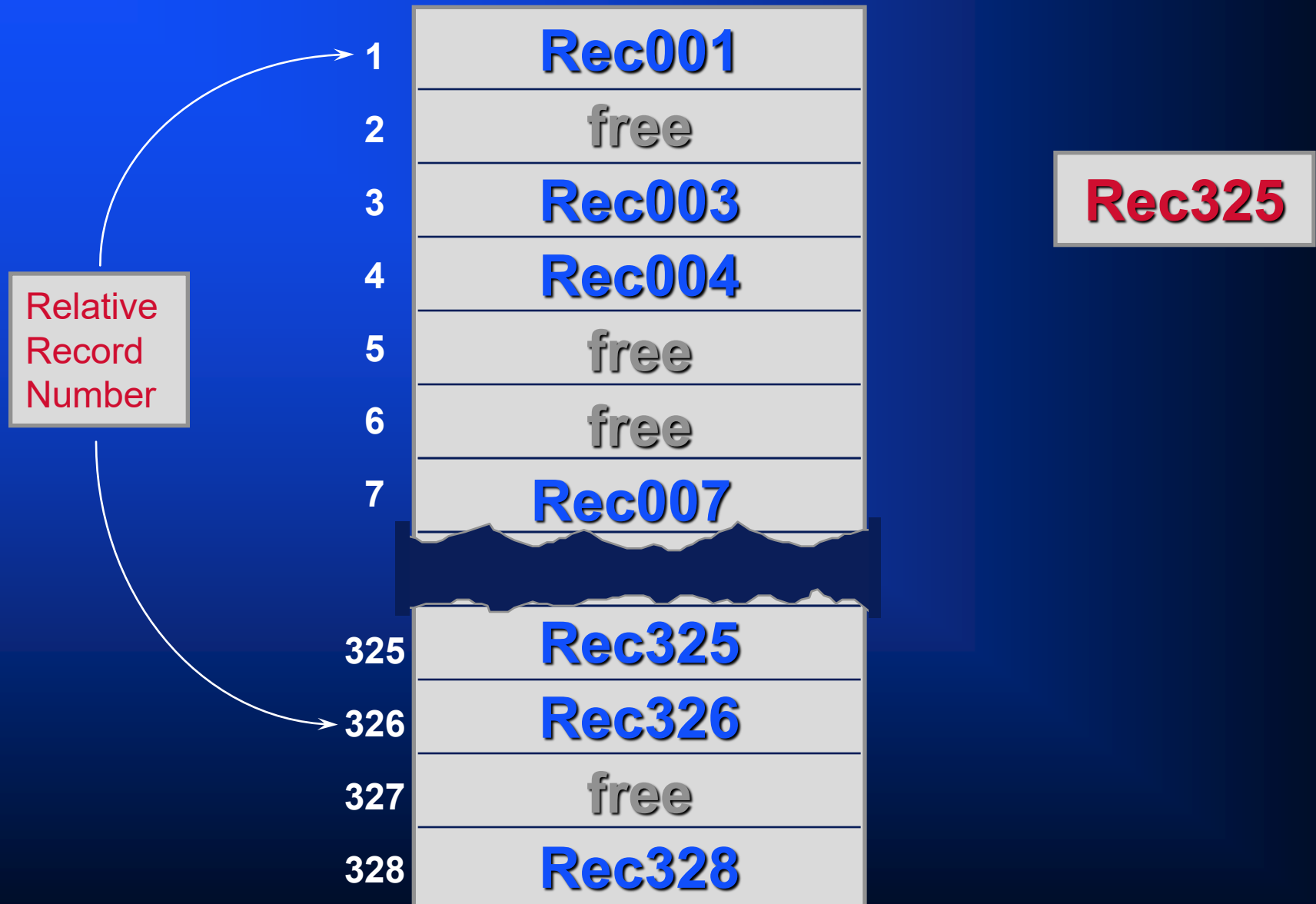
Relative Files - Adding a Record



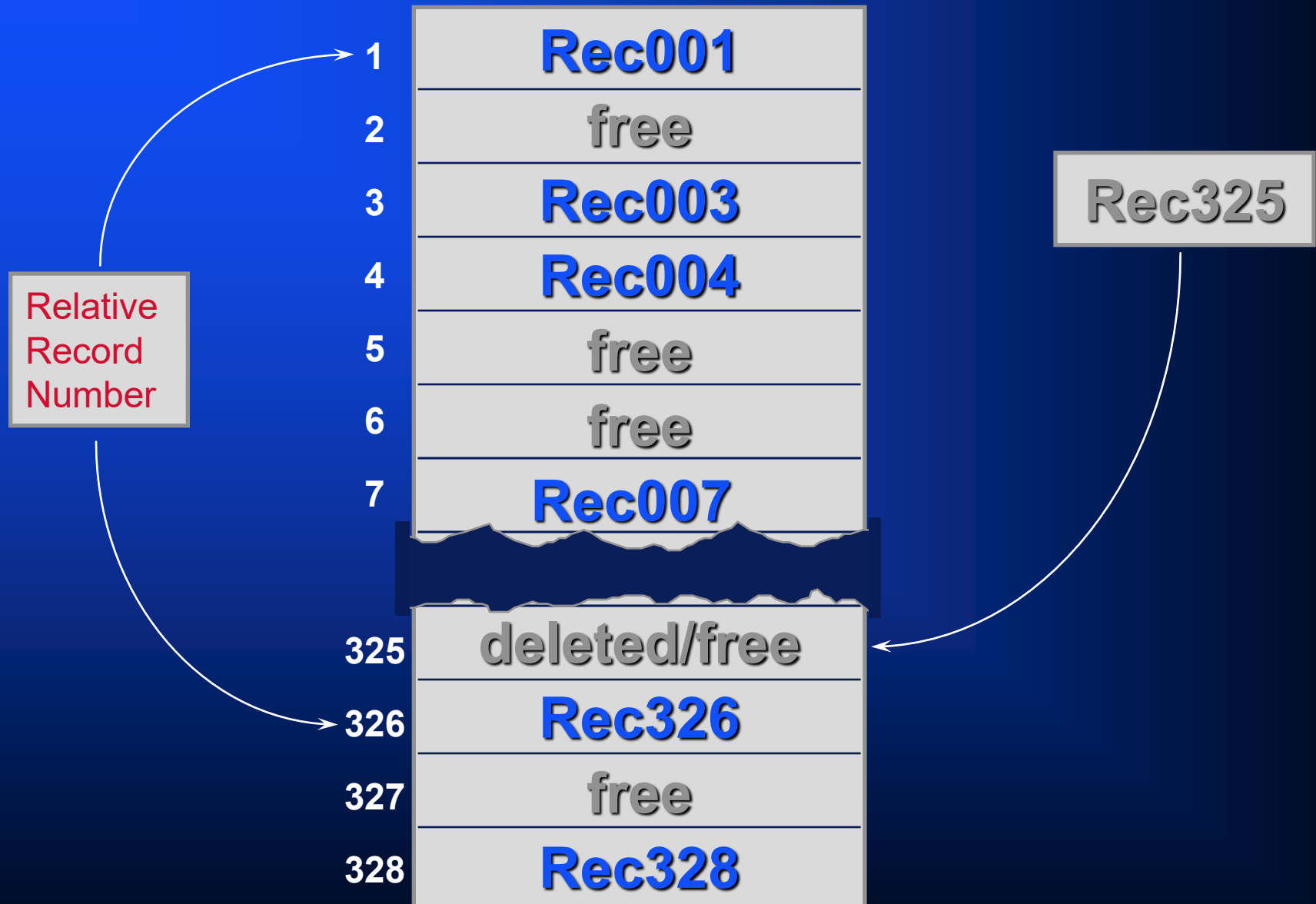
Relative Files - Adding a Record



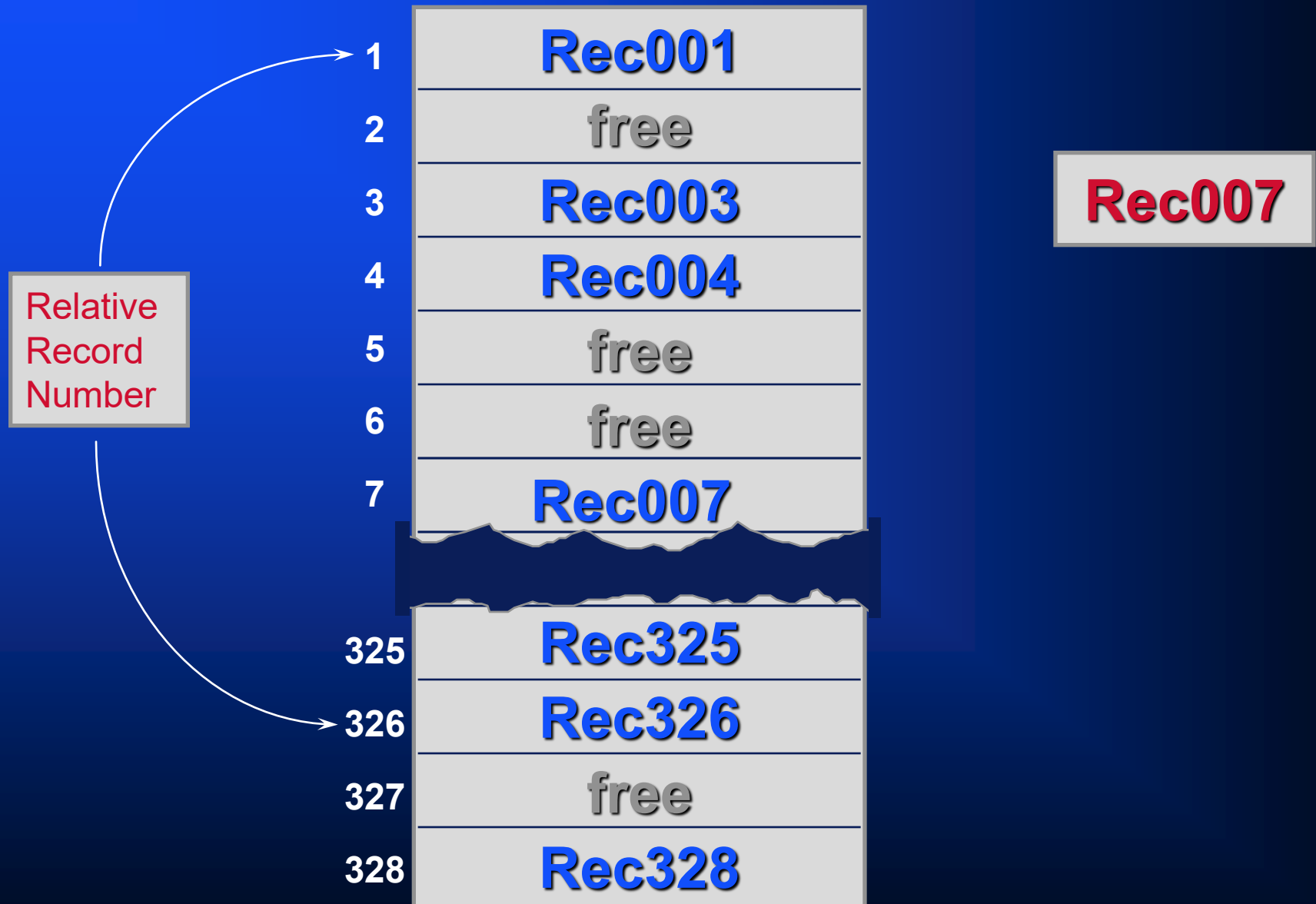
Relative Files - Deleting a Record



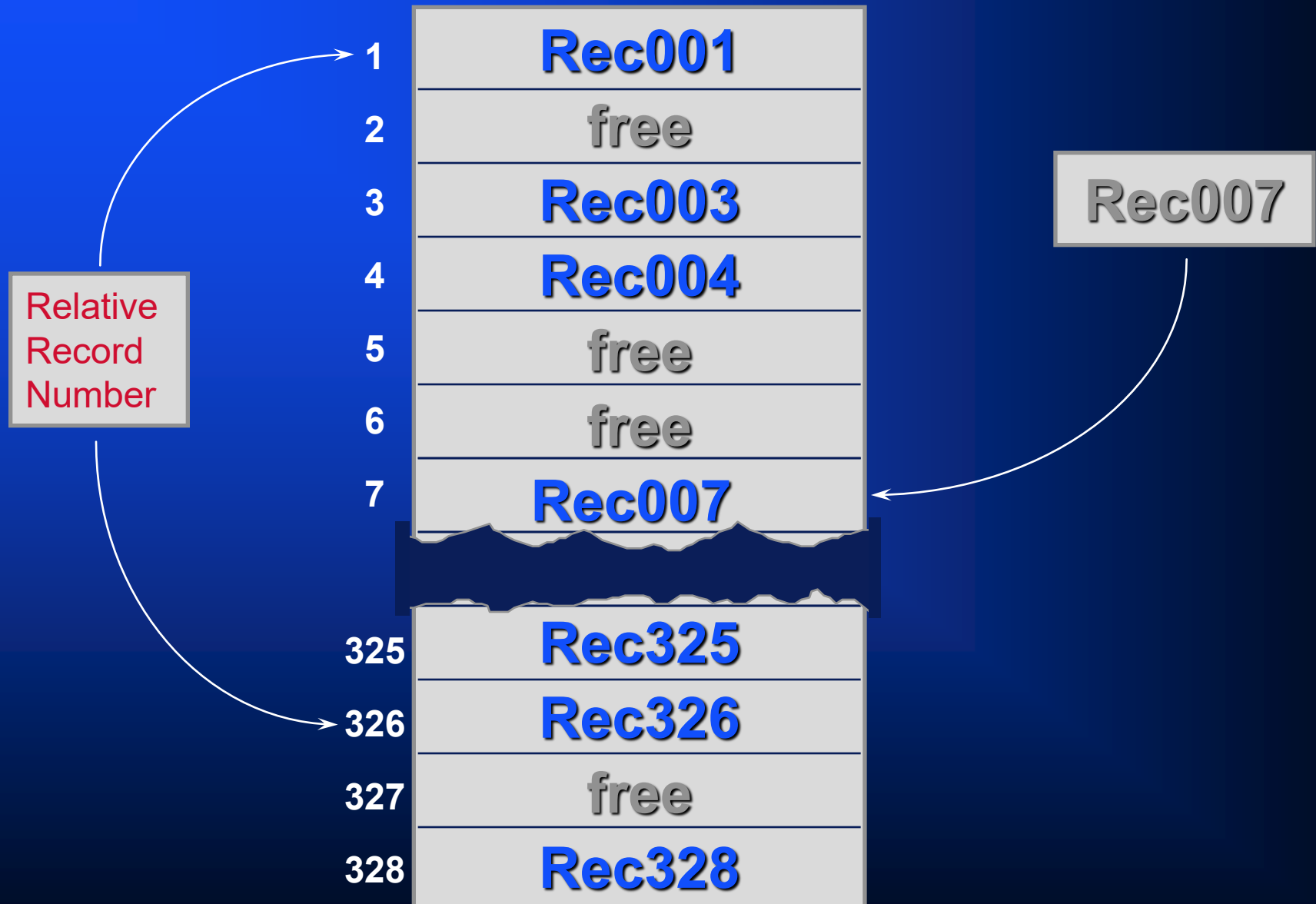
Relative Files - Deleting a Record



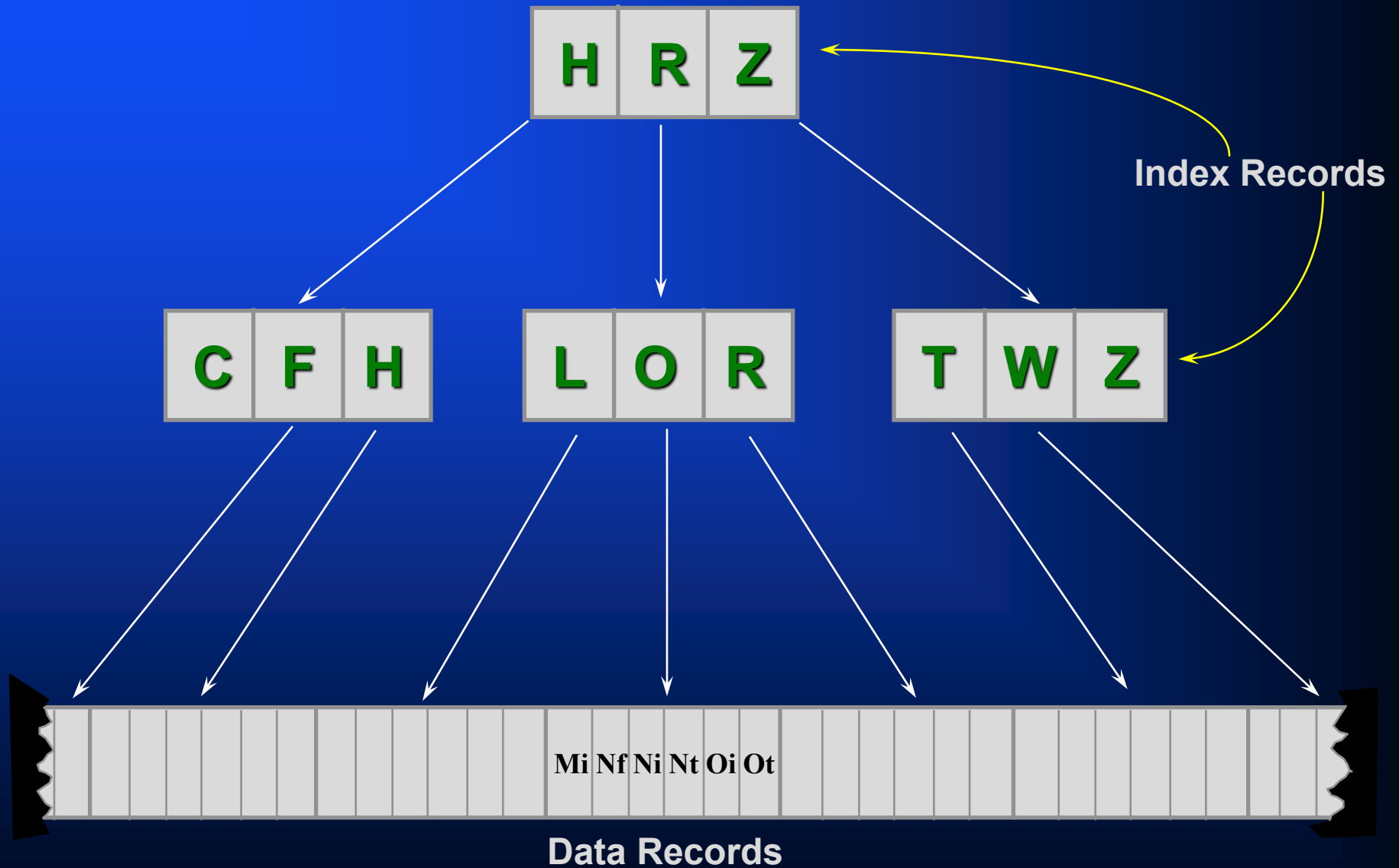
Relative Files - Amending a Record



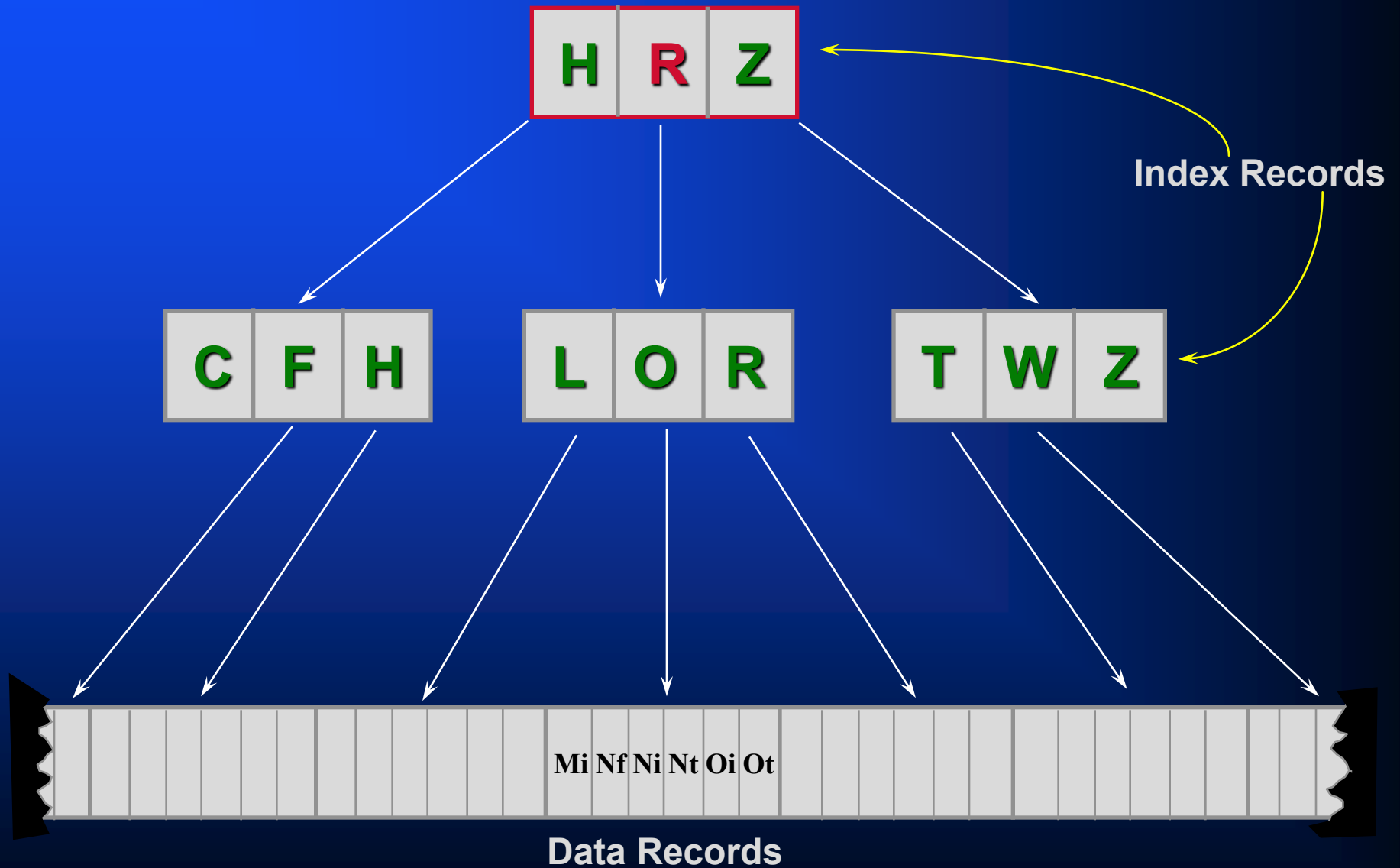
Relative Files - Amending a Record



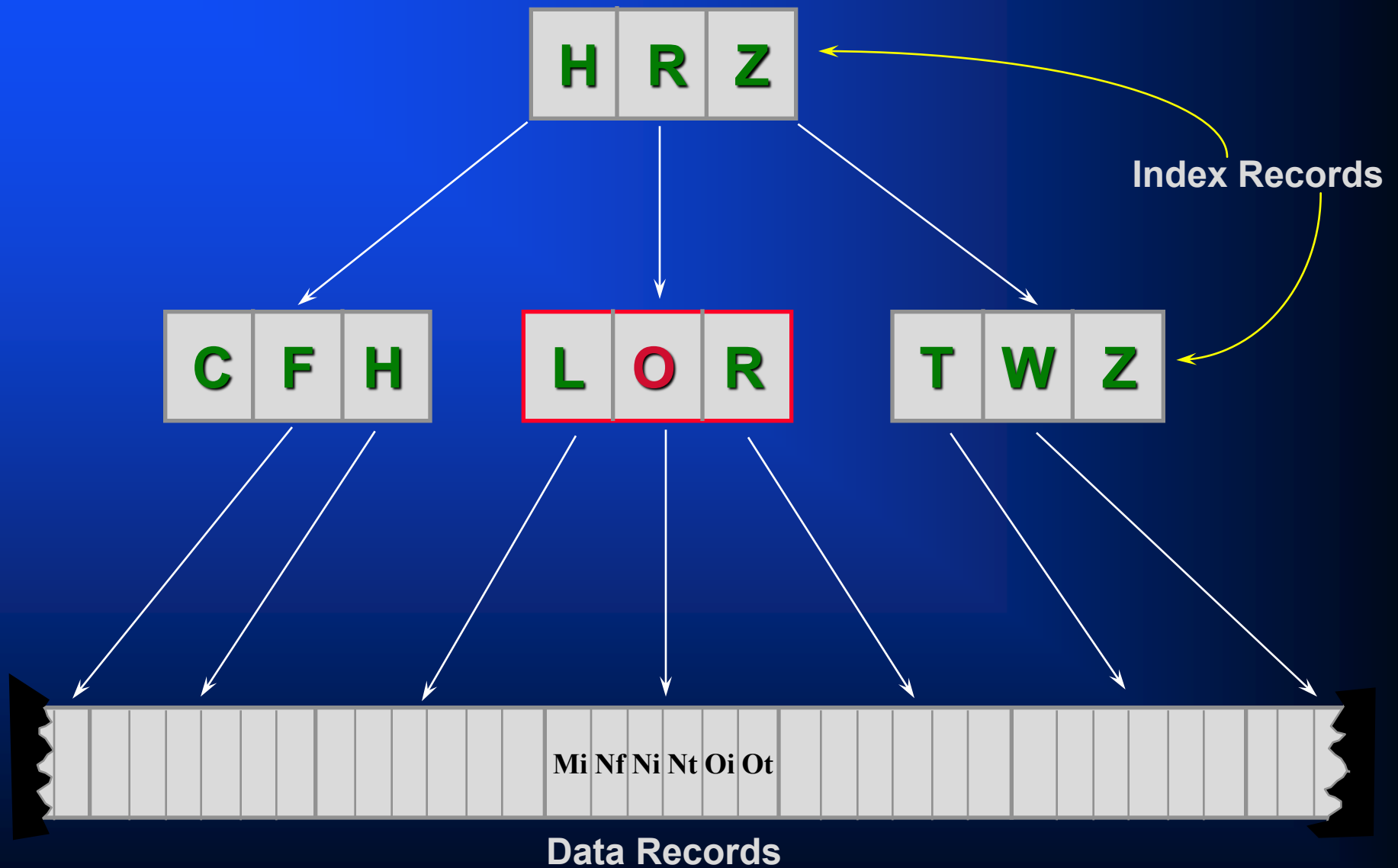
Indexed Files - Organization



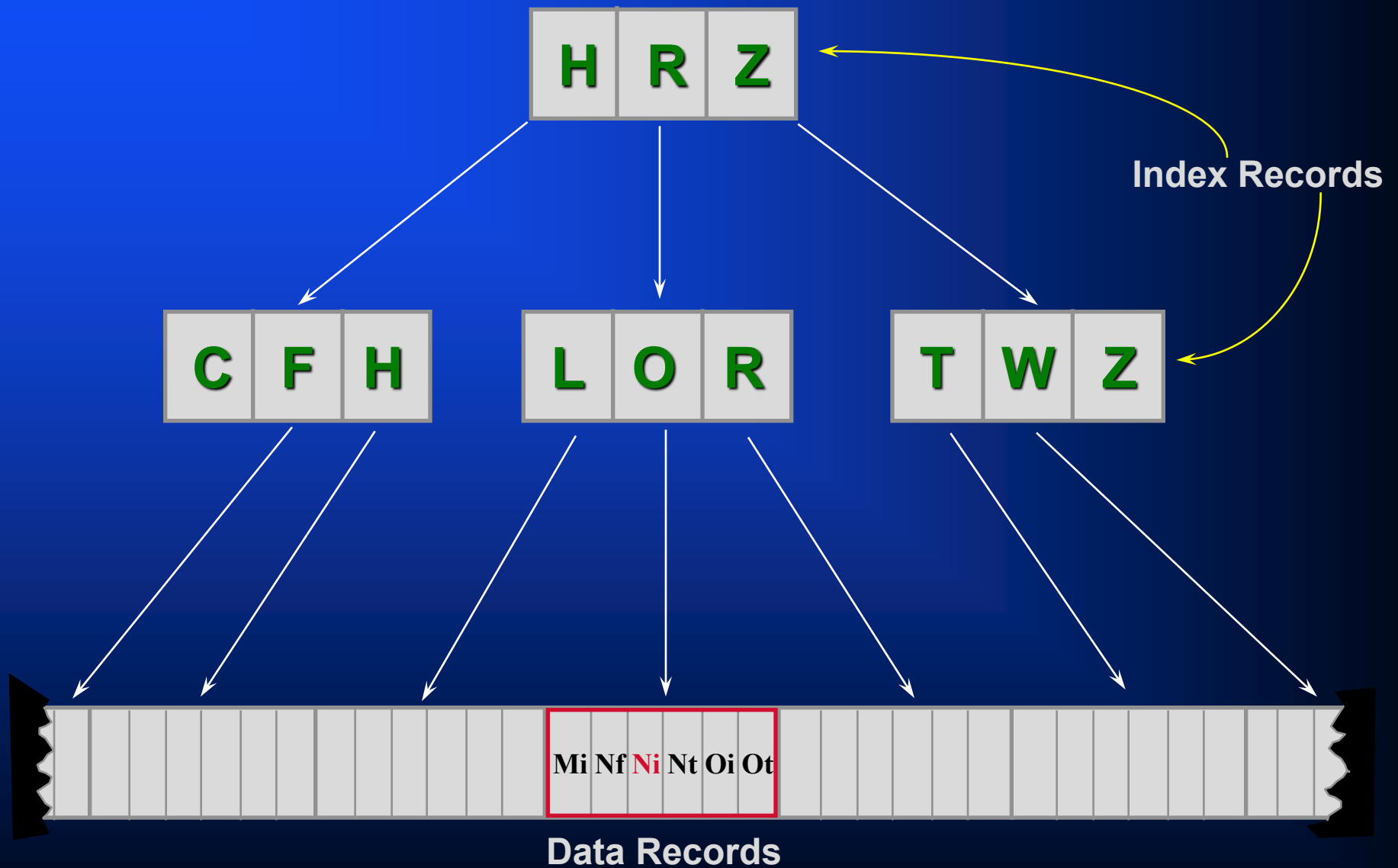
Indexed Files - Reading Record N_i



Indexed Files - Reading Record N_i



Indexed Files - Reading Record N_i



Sequential Files.

Disadvantages.

- ◆ **Slow** - when the hit rate is low.
- ◆ **Complicated** to change (insert, delete, amend)

Advantages.

- ◆ **Fast** - when the hit rate is high.
- ◆ Most **storage efficient**.
- ◆ **Simple** organization.
- ◆ **Recovers space** from deleted records.



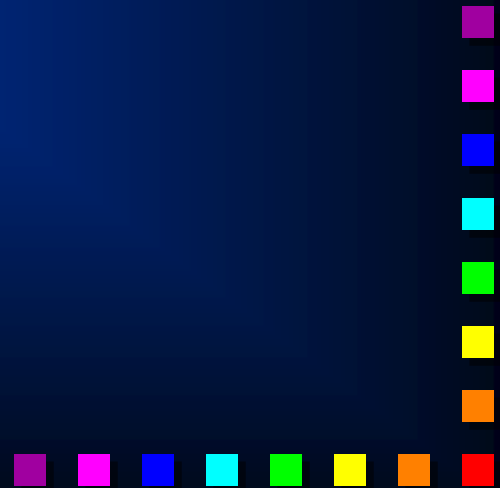
Relative Files.

Disadvantages.

- ◆ **Wasteful** of storage if the file is only partially populated.
- ◆ **Cannot recover space** from deleted records.
- ◆ Only a **single, numeric key** allowed.
- ◆ Keys must map on to the range of the Relative Record numbers.

Advantages.

- ◆ **Fastest** Direct Access organization.
- ◆ Very **little storage overhead**.
- ◆ Can be read sequentially.



Indexed Files.

Disadvantages.

- ◆ **Slowest** Direct Access organization.
- ◆ **Especially slow** when adding or deleting records.
- ◆ **Not very storage efficient.** Must store the Index records, the alternate Index records, the data records and the alternate data records.

Advantages.

- ◆ Can use **multiple, alphanumeric keys.**
- ◆ Can have **duplicate alternate keys.**
- ◆ Can be **read sequentially** on any of its keys.
- ◆ Can **partially recover space** from deleted records.

